EE114 - Homework 1

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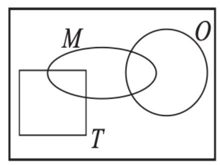
Quiz 1.1

A pizza at Gerlanda's is either regular (R) or Tuscan (T). In addition, each slice may have mushrooms (M) or onions (O) as described by the Venn diagram. For the sets specified below, shade the corresponding region of the Venn diagram.



(1) R
Solution: Shade the circle R.

(2) $M \cup O$
Solution: Shade the union of circles M and O.

(3) $M \cap O$
Solution: Shade the intersection of circles M and O.

(4) $R \cup M$
Solution: Shade values in either R or M (their union).

(5) $R \cap M$
Solution: Shade the intersection of R and M.

(6) $T^c - M$
Solution: T^c represents everything not in T, which corresponds to R. Then subtract M. Shade the region of R that does not overlap with M.

Problem 1.1.1

For Gerlanda's pizza in Quiz 1.1, answer these questions:

(a) Are T and M mutually exclusive?
Solution: No. A Tuscan pizza can have mushrooms. The intersection $T \cap M$ is not empty.

(b) Are R, T, and M collectively exhaustive?
Solution: Yes. Since pizza types are $\{R, T\}$, $R \cup T = S$. Thus $R \cup T \cup M = S$. They cover the entire sample space.

(c) Are T and O mutually exclusive? State this condition in words.
Solution: No. A Tuscan pizza can have onions. Condition: $T \cap O = \emptyset$ (false here).

(d) Does Gerlanda's make Tuscan pizzas with mushrooms and onions?
Solution: Yes, provided the intersection $T \cap M \cap O$ is non-empty.

(e) Does Gerlanda's make regular pizzas that have neither mushrooms nor onions?
Solution: Yes, this is the region $R \cap M^c \cap O^c$.

Problem 1.2.1

A fax transmission can take place at any of three speeds depending on the condition of the phone connection between the two fax machines. The speeds are high (h) at 14,400 b/s, medium (m) at 9600 b/s, and low (l) at 4800 b/s. In response to requests for information, a company sends either short faxes of two (t) pages, or long faxes of four (f) pages. Consider the experiment of monitoring a fax transmission and observing the transmission speed and length. An observation is a two-letter word, for example, a high-speed, two-page fax is ht.

(a) What is the sample space of the experiment?
Solution: $S = \{ht, hf, mt, mf, lt, lf\}$.

(b) Let A_1 be the event "medium-speed fax." What are the outcomes in A_1?
Solution: $A_1 = \{mt, mf\}$.

(c) Let A_2 be the event "short (two-page) fax." What are the outcomes in A_2?
Solution: $A_2 = \{ht, mt, lt\}$.

(d) Let A_3 be the event "high-speed fax or low-speed fax." What are the outcomes in A_3?
Solution: $A_3 = \{ht, hf, lt, lf\}$.

(e) Are A_1, A_2, and A_3 mutually exclusive?
Solution: No. $A_1 \cap A_2 = \{mt\} \neq \emptyset$.

(f) Are A_1, A_2, and A_3 collectively exhaustive?
Solution: Yes. $A_1 \cup A_2 \cup A_3 = \{mt, mf, ht, lt, hf, lf\} = S$.

Problem 1.2.2

An integrated circuit factory has three machines X, Y, and Z. Test one integrated circuit produced by each machine. Either a circuit is acceptable (a) or it fails (f). An observation is a sequence of three test results corresponding to the circuits from machines X, Y, and Z, respectively. For example, aaf is the observation that the circuits from X and Y pass the test and the circuit from Z fails the test.

(a) What are the elements of the sample space of this experiment?
Solution: $S = \{aaa, aaf, afa, aff, faa, faf, ffa, fff\}$.

(b) What are the elements of the sets

$$Z_F = \{\text{circuit from } Z \text{ fails}\},$$

$$X_A = \{\text{circuit from } X \text{ is acceptable}\}.$$

Solution:
$Z_F = \{aaf, aff, faf, fff\}$
$X_A = \{aaa, aaf, afa, aff\}$

(c) Are Z_F and X_A mutually exclusive?
Solution: No. $Z_F \cap X_A = \{aaf, aff\} \neq \emptyset$.

(d) Are Z_F and X_A collectively exhaustive?
Solution: No. $Z_F \cup X_A = S \setminus \{faa, ffa\}$. It is missing elements where X fails and Z passes.

(e) What are the elements of the sets

$$C = \{\text{more than one circuit acceptable}\},$$

$$D = \{\text{at least two circuits fail}\}.$$

Solution:
$C = \{2 \text{ or } 3 \text{ 'a's}\} = \{aaa, aaf, afa, faa\}$
$D = \{2 \text{ or } 3 \text{ 'f's}\} = \{aff, faf, ffa, fff\}$

(f) Are C and D mutually exclusive?
Solution: Yes. C contains outcomes with ≥ 2 a's (thus ≤ 1 f), D contains outcomes with ≥ 2 f's. They cannot overlap.

(g) Are C and D collectively exhaustive?
Solution: Yes. Any outcome has either ≤ 1 f (in C) or ≥ 2 f (in D). They partition the space.

Problem 1.3.1

Computer programs are classified by the length of the source code and by the execution time. Programs with more than 150 lines in the source code are big (B). Programs with ≤ 150 lines are little (L). Fast programs (F) run in less than 0.1 seconds. Slow programs (W) require at least 0.1 seconds. Monitor a program executed by a computer. Observe the length of the source code and the run time. The probability model for this experiment contains the following information: $P[LF] = 0.5$, $P[BF] = 0.2$, and $P[BW] = 0.2$. What is the sample space of the experiment? Calculate the following probabilities:

(a) $P[W]$
Solution: Given $P[LF] = 0.5, P[BF] = 0.2, P[BW] = 0.2$. Sum of elementary events is 1.
$P[LF] + P[LW] + P[BF] + P[BW] = 1 \implies 0.5 + P[LW] + 0.2 + 0.2 = 1 \implies P[LW] = 0.1$.
$P[W] = P[LW] + P[BW] = 0.1 + 0.2 = 0.3$.

(b) $P[B]$
Solution: $P[B] = P[BF] + P[BW] = 0.2 + 0.2 = 0.4$.

(c) $P[W \cup B]$
Solution: $P[W \cup B] = P[W] + P[B] - P[W \cap B]$. $W \cap B = BW$.
So $P[W \cup B] = 0.3 + 0.4 - 0.2 = 0.5$.

Problem 1.3.2

There are two types of cellular phones, handheld phones (H) that you carry and mobile phones (M) that are mounted in vehicles. Phone calls can be classified by the traveling speed of the user as fast (F) or slow (W). Monitor a cellular phone call and observe the type of telephone and the speed of the user. The probability model for this experiment has the following information: $P[F] = 0.5$, $P[HF] = 0.2$, $P[MW] = 0.1$. What is the sample space of the experiment? Calculate the following probabilities:

(a) $P[W]$
Solution: $P[W] = 1 - P[F] = 1 - 0.5 = 0.5$.

(b) $P[MF]$
Solution: $P[F] = P[HF] + P[MF] \implies 0.5 = 0.2 + P[MF] \implies P[MF] = 0.3$.

(c) $P[H]$
Solution: First find $P[M]$. $P[M] = P[MF] + P[MW] = 0.3 + 0.1 = 0.4$.
Then $P[H] = 1 - P[M] = 1 - 0.4 = 0.6$. (Check: $P[H] = P[HF] + (P[W] - P[MW]) = 0.2 + (0.5 - 0.1) = 0.6$).

Problem 1.4.1

Mobile telephones perform *handoffs* as they move from cell to cell. During a call, a telephone either performs zero handoffs (H_0), one handoff (H_1), or more than one handoff (H_2). In addition, each call is either long (L), if it lasts more than three minutes, or brief (B). The following table describes the probabilities of the possible types of calls.

	H_0	H_1	H_2
L	0.1	0.1	0.2
B	0.4	0.1	0.1

What is the probability $P[H_0]$ that a phone makes no handoffs?
Solution: $P[H_0] = 0.1 + 0.4 = 0.5$.
What is the probability a call is brief?
Solution: $P[B] = 0.4 + 0.1 + 0.1 = 0.6$.
What is the probability a call is long or there are at least two handoffs?
Solution: $P[L \cup H_2] = P[L] + P[H_2] - P[L \cap H_2]$.
$P[L] = 0.1 + 0.1 + 0.2 = 0.4$.
$P[H_2] = 0.2 + 0.1 = 0.3$.
$P[L \cap H_2] = 0.2$.
$P[L \cup H_2] = 0.4 + 0.3 - 0.2 = 0.5$.

Problem 1.4.2

For the telephone usage model of Example 1.14, let B_m denote the event that a call is billed for m minutes. To generate a phone bill, observe the duration of the call in integer minutes (rounding up). Charge for M minutes $M = 1, 2, 3, \ldots$ if the exact duration T is $M - 1 < t \leq M$. A more complete probability model shows that for $m = 1, 2, \ldots$ the probability of each event B_m is

$$P[B_m] = \alpha(1-\alpha)^{m-1}$$

where $\alpha = 1 - (0.57)^{1/3} = 0.171$.

(a) Classify a call as long, L, if the call lasts more than three minutes. What is $P[L]$?
Solution: A call lasts > 3 minutes if it is billed for $m \geq 4$.
$P[L] = \sum_{m=4}^{\infty} P[B_m] = \sum_{m=4}^{\infty} \alpha(1-\alpha)^{m-1}$.
Let $k = m - 1$, sum is $\sum_{k=3}^{\infty} \alpha(1-\alpha)^k = \frac{\alpha(1-\alpha)^3}{1-(1-\alpha)} = (1-\alpha)^3$.
Given $\alpha = 1 - 0.57^{1/3}$, we have $1 - \alpha = 0.57^{1/3}$.
So $P[L] = (0.57^{1/3})^3 = 0.57$.

(b) What is the probability that a call will be billed for nine minutes or less?
Solution: $P[M \leq 9] = 1 - P[M > 9]$.
$M > 9 \implies M \geq 10$.
$P[M \geq 10] = \sum_{m=10}^{\infty} P[B_m] = (1-\alpha)^9$.
Using $1 - \alpha = 0.57^{1/3}$, $P[M \geq 10] = (0.57^{1/3})^9 = 0.57^3 \approx 0.185$.
$P[M \leq 9] = 1 - 0.185 = 0.815$.

Problem 1.4.3

The basic rules of genetics were discovered in mid-1800s by Mendel, who found that each characteristic of a pea plant, such as whether the seeds were green or yellow, is determined by two genes, one from each parent. Each gene is either dominant d or recessive r. Mendel's experiment is to select a plant and observe whether the genes are both dominant d, both recessive r, or one of each (hybrid) h. In his pea plants, Mendel found that yellow seeds were a dominant trait over green seeds. A yy pea with two yellow genes has yellow seeds; a gg pea with two recessive genes has green seeds; a hybrid gy or yg pea has yellow seeds. In one of Mendel's experiments, he started with a parental generation in which half the pea plants were yy and half the plants were gg. The two groups were crossbred so that each pea plant in the first generation was gy. In the second generation, each pea plant was equally likely to inherit a y or a g gene from each first generation parent. What is the probability $P[Y]$ that a randomly chosen pea plant in the second generation has yellow seeds?

Solution:

Start: $P(yy) = 0.5, P(gg) = 0.5$.

F1: All gy.

F2: Cross $gy \times gy$.

Offspring genotypes: yy, yg, gy, gg each with probability 1/4.

Yellow seeds occur for yy, yg, gy.

$P[Y] = P(yy) + P(yg) + P(gy) = 1/4 + 1/4 + 1/4 = 3/4 = 0.75$.

Problem 1.4.4

Use Theorem 1.7 to prove the following facts:

(a) $P[A \cup B] \geq P[A]$
Proof: $A \subseteq A \cup B$. By Monotonicity Axiom, $P[A] \leq P[A \cup B]$.

(b) $P[A \cup B] \geq P[B]$
Proof: $B \subseteq A \cup B$. By Monotonicity Axiom, $P[B] \leq P[A \cup B]$.

(c) $P[A \cap B] \leq P[A]$
Proof: $A \cap B \subseteq A$. By Monotonicity Axiom, $P[A \cap B] \leq P[A]$.

(d) $P[A \cap B] \leq P[B]$
Proof: $A \cap B \subseteq B$. By Monotonicity Axiom, $P[A \cap B] \leq P[B]$.

Problem 1.4.5

Use Theorem 1.7 to prove by induction the *union bound*: For any collection of events $A_1, \ldots, A_n$,

$$P[A_1 \cup A_2 \cup \cdots \cup A_n] \leq \sum_{i=1}^{n} P[A_i].$$

Proof:
Base Case ($n = 2$): $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$. Since $P(A_1 \cap A_2) \geq 0$, $P(A_1 \cup A_2) \leq P(A_1) + P(A_2)$.
Induction Step: Assume true for $n = k$.
$P(\cup_{i=1}^{k+1} A_i) = P((\cup_{i=1}^{k} A_i) \cup A_{k+1})$.
By base case, this is $\leq P(\cup_{i=1}^{k} A_i) + P(A_{k+1})$.
By inductive hypothesis, $P(\cup_{i=1}^{k} A_i) \leq \sum_{i=1}^{k} P(A_i)$.
Substituting this in, $P(\cup_{i=1}^{k+1} A_i) \leq \sum_{i=1}^{k} P(A_i) + P(A_{k+1}) = \sum_{i=1}^{k+1} P(A_i)$.
The statement holds for all n.